

# pktPCAP — Network & Security Analyst Guide

[Packet Capture Analysis Platform](#) · [Analyst Reference](#)

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## Overview

pktPCAP is a browser-based packet capture analysis platform. Analysts upload `.pcap`, `.pcapng`, or `.cap` files — or stream live capture feeds from remote hosts — and investigate traffic through an interactive multi-tab dashboard with an AI-assisted chat engine.

This guide covers everything you need to use pktPCAP day-to-day: logging in, uploading captures, connecting live feeds, navigating the analysis tabs, working with the AI engine, and exporting results.

# Logging In

Open your organization's pktPCAP URL in any modern browser. You will see the login screen.

## Local Authentication

Enter your assigned **username** and **password**, then click **Sign In**. Contact your platform administrator if you have not yet been issued credentials.

## Single Sign-On (SSO)

If your organization has configured SAML SSO, click **Sign in with SSO**. You will be redirected to your identity provider. After authenticating there, you return to pktPCAP automatically. SSO accounts are provisioned on first login — no pre-registration required.

**Note:** Your username and assigned role (Admin / Analyst / Viewer) appear in the sidebar footer after login. Your role determines which features are available to you.

# The Interface

The pktPCAP interface has four primary areas:

| Area                  | Description                                                         |
|-----------------------|---------------------------------------------------------------------|
| Left Sidebar          | Navigation between Dashboard, Live Feeds, and Settings (Admin only) |
| Upload / Landing Zone | Central drop zone when no capture is loaded — drag files here       |
| Data Panel            | Tabbed analysis pane with all protocol, stream, and endpoint views  |
| AI Chat Panel         | Right-side conversation panel for querying the AI engine            |

Click ↺ **New Analysis** in the top bar at any time to clear the current session and start fresh.

# Uploading PCAP Files

Uploading a capture file is the primary way to begin an analysis session.

## Supported Formats

- `.pcap` — Standard libpcap format (Wireshark, tcpdump)
- `.pcapng` — Next-generation pcap format
- `.cap` — Capture files from various network tools

## Upload Steps

- Step 1 — Select your file(s)** Drag one or more files directly onto the upload drop zone, or click the zone to open a file picker. Multiple files can be selected at once.
- Step 2 — Multi-file merge** When multiple files are uploaded together, pktPCAP automatically merges and correlates them into a unified session. This is useful for analyzing captures from multiple taps or time windows simultaneously.
- Step 3 — Start Analysis** Click **Analyze**. The server parses the capture, decodes all layers, reconstructs streams, and resolves DNS. The data panel and AI panel load when processing completes.

***Tip:** Very large captures (several hundred MB or more) may take additional processing time. Consider pre-filtering with Wireshark or tshark to focus on the relevant timeframe before uploading.*

# Live Feed Capture

Live feeds stream packet data from a remote host directly into pktPCAP in real time — no intermediate file required. Navigate to **Live Feeds** in the sidebar.

## Connecting a Remote Host

Fill in the connection form:

| Field     | Description                                                    |
|-----------|----------------------------------------------------------------|
| Feed Name | A label for this session (e.g., core-switch-uplink)            |
| Interface | The network interface to capture on (e.g., eth0, ens3)         |
| Filter    | Optional BPF filter (e.g., not port 22 to exclude SSH traffic) |

## tshark (CLI) Method

After configuring the options, pktPCAP generates a tshark command to run on your remote host. Click **Copy Command** and execute it in an SSH session on the target machine. The feed appears in pktPCAP within seconds.

## Wireshark (GUI) Method

Select the **Wireshark** tab in the Live Feeds section. Configure Wireshark to use pktPCAP as its remote capture destination. The stream is simultaneously displayed in Wireshark and recorded in pktPCAP.

***Security Note:** Each live feed has a unique authentication token embedded in the generated command. Treat it like a password — do not share it outside the capture workflow.*

# Analysis Tab Reference

After a capture is loaded, the **Data Panel** contains multiple tabs. Each surfaces a different view of the capture data.

| Tab           | What It Shows                                                    |
|---------------|------------------------------------------------------------------|
| Summary       | Packet counts, byte totals, capture duration, top protocols      |
| Protocols     | Hierarchical protocol breakdown with counts and byte totals      |
| Streams       | Reconstructed TCP/UDP conversations with full payload access     |
| Endpoints     | All communicating IPs: packet/byte counts, resolved hostnames    |
| Conversations | Bi-directional traffic pairs across all transport protocols      |
| Packets       | Frame-level view with timestamps, addresses, protocol, info      |
| DNS           | All DNS queries and responses with record types and results      |
| TLS / HTTP    | TLS handshake details (SNI, cipher suites) and HTTP transactions |

## Using the Streams Tab

Click any stream row to load the full payload sequence. This is most useful for:

- Inspecting HTTP request/response cycles
- Following DNS lookup chains
- Reviewing unencrypted application protocol exchanges

## Using the Endpoints Tab

Sort by any column (packets, bytes, hostname) to surface:

- Top talkers by volume
- Unexpected external communicators
- Hosts generating unusual connection counts

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# AI-Assisted Analysis

The **Chat Panel** on the right connects to an AI engine (OpenAI or Anthropic Claude). After analysis runs, the AI automatically generates an opening summary of the capture. You can continue the conversation with natural-language questions.

## What the AI Can Help With

- Summarizing traffic patterns and anomalies in plain language
- Identifying unusual port usage, connection failures, or protocol mismatches
- Explaining what a specific protocol exchange means
- Correlating events across streams and endpoints
- Flagging potential indicators of compromise from traffic patterns
- Answering questions about data volumes, top talkers, and session behavior

## Example Prompts

"What are the top 5 external IPs this host is communicating with?"

"Are there any failed authentication attempts or reset connections in this capture?"

"Summarize the DNS queries — are there any unusual domains being resolved?"

"Which TLS versions are in use? Are any sessions negotiating outdated cipher suites?"

"Is there any evidence of data exfiltration or unusually large outbound transfers?"

**Privacy Note:** Raw packet payload bytes are **never** sent to the AI provider. Only structured metadata, protocol summaries, and your typed questions are transmitted. All raw capture data remains on your organization's server.

## Exporting Results

Analysis findings can be exported for use in incident tickets, reports, or knowledge base articles.

### Available Exports

| Export           | How to Access                                |
|------------------|----------------------------------------------|
| Protocol Summary | Export button in the Protocols tab           |
| Endpoint List    | CSV-compatible export from the Endpoints tab |
| AI Analysis Text | Copy the full chat thread from the AI panel  |
| Stream Payloads  | Copy directly from the Streams tab           |

## Dashboard Screenshots

The **Screenshot** function captures the current dashboard state — including the data panel and AI chat — as an image file saved on the server. This is useful for visual record-keeping or including UI state in documentation.

*If the screenshot feature is not responding, contact your administrator to verify that server-side screenshot tooling is installed and configured.*

## Tips for Effective Analysis

1. **Start with Summary** — always review the top-level statistics before diving into specific tabs. Anomalies in packet counts or protocol distribution often point you to the right area.
2. **Use BPF filters for live feeds** — capture only what you need. `host 10.x.x.x` or `port 443` dramatically reduces noise for targeted investigations.
3. **Merge multiple captures** — if you have captures from multiple sensors covering the same incident window, upload them together. pktPCAP correlates them into a unified timeline.
4. **Ask the AI specific questions** — the more specific your question, the more targeted the AI response. "Show me all RST packets" is better than "what's interesting?"
5. **Export early** — export AI summaries and endpoint lists before starting a new analysis session, as loading a new capture replaces the current session view.